# Too Hot To Be True:
## Temperature Calibration for Higher Confidence in NN-assisted SCA

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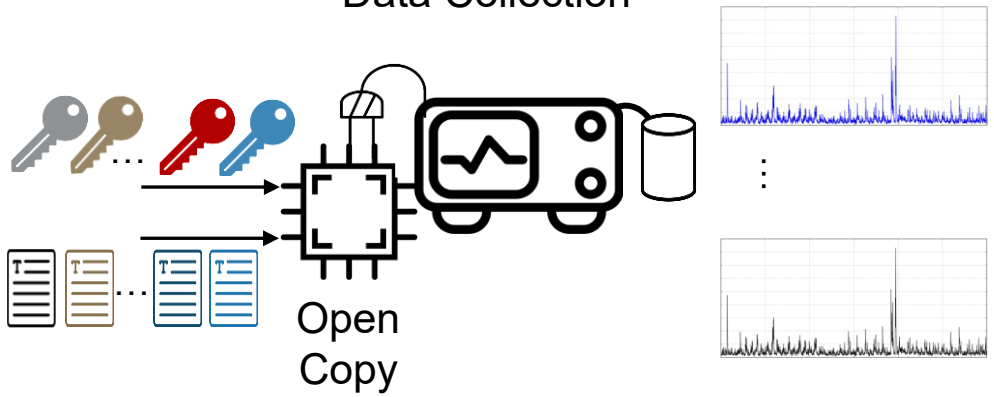
CASCADE'26

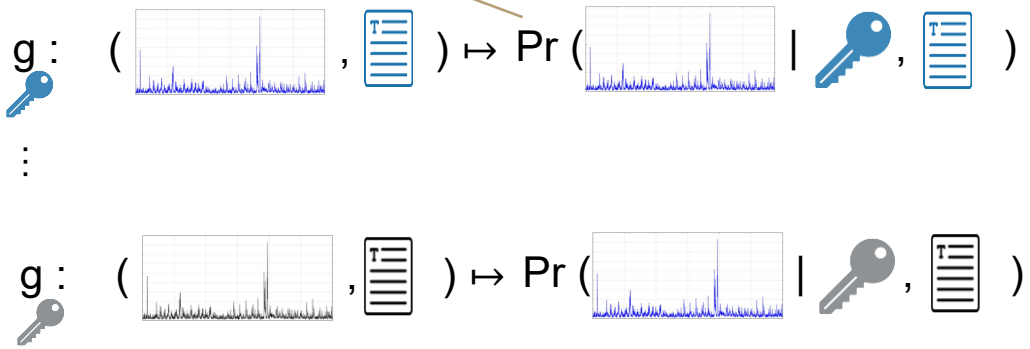
# Profiled SCA: in search for probability distributions!

Prob. Density Function

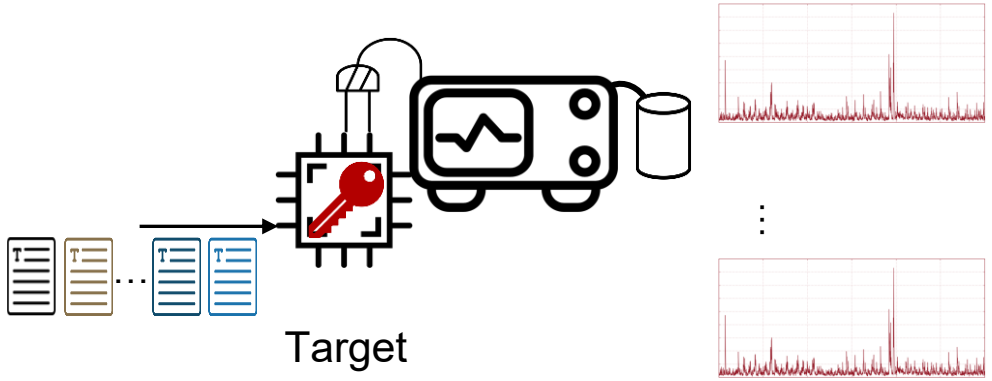
Profiling

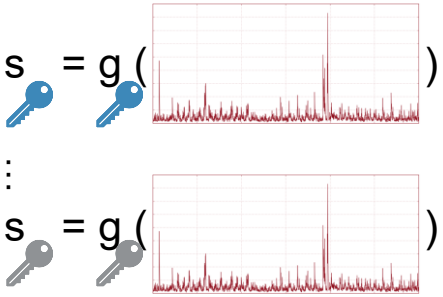
Data Collection

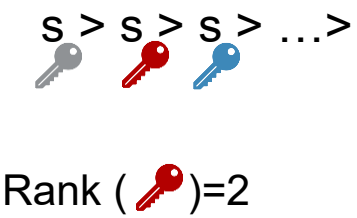



Attack

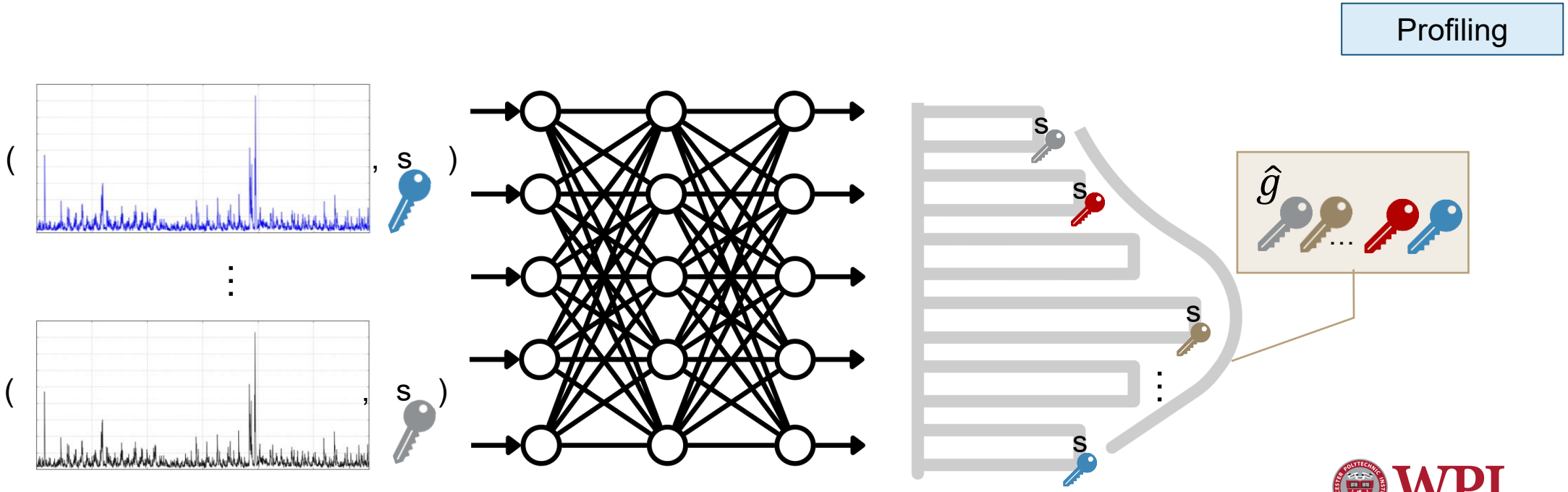


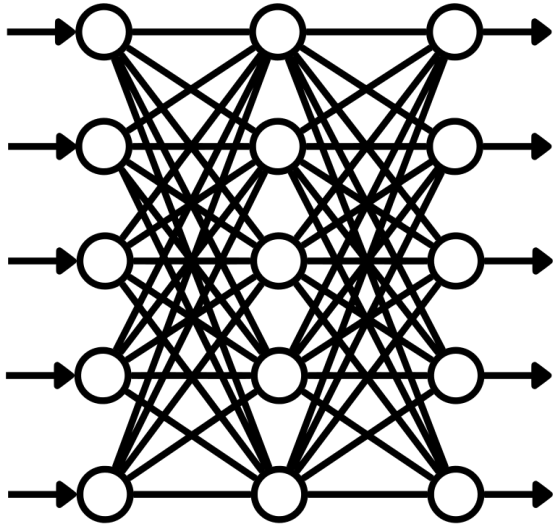
# Let’s bring in NNs

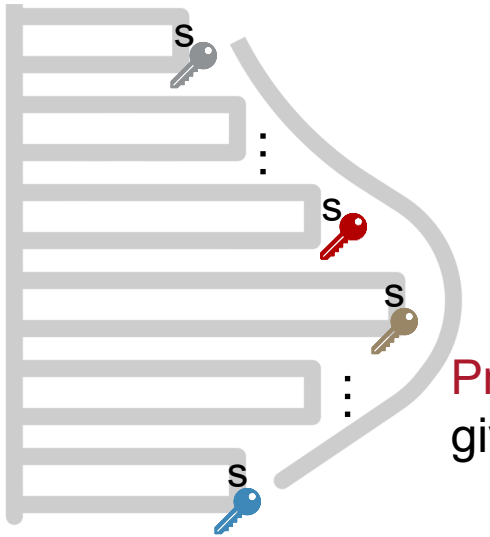
- Characterizing the leakage precisely through statistical techniques: costly in terms of the number of traces needed
- NN-assisted profiled SCA: effective against un-/protected cryptographic implementations, as well as noisy and shuffled traces



# NN-assisted SCA

Attack






Rank ( )=10

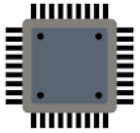
Problem: a high prediction probability given to an incorrect key

NN model is too confident in its predictions

Confidence = Pr ( is the correct key)

# How does it look like in practice?

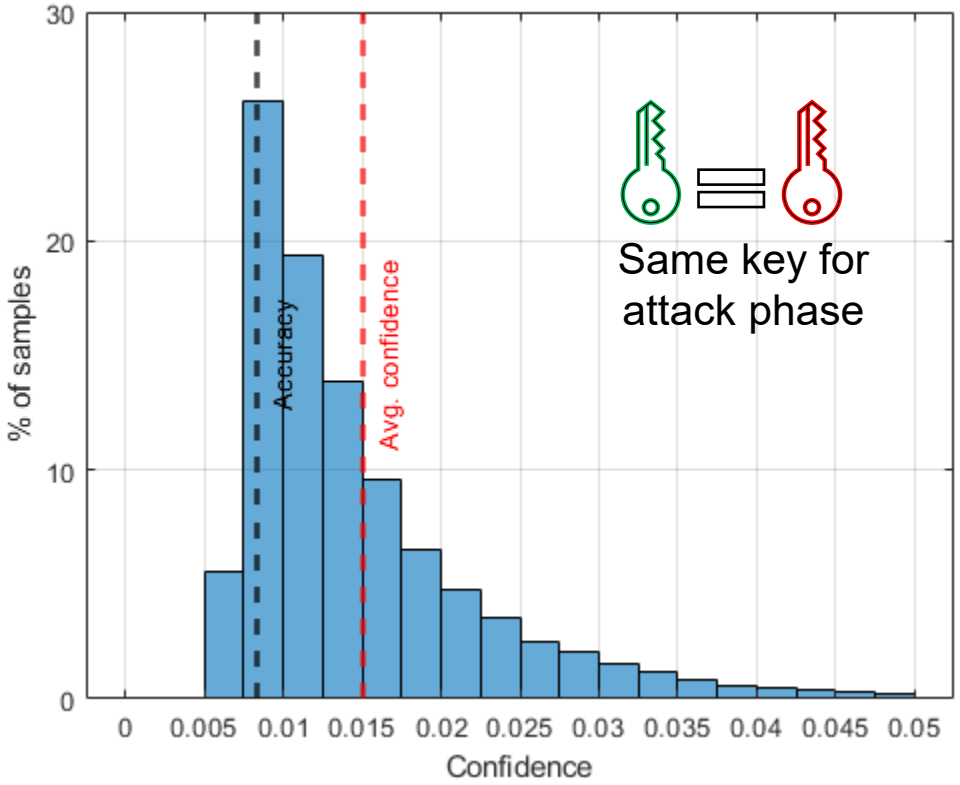
- ASCAD dataset [1]
- Models proposed in [2,3]

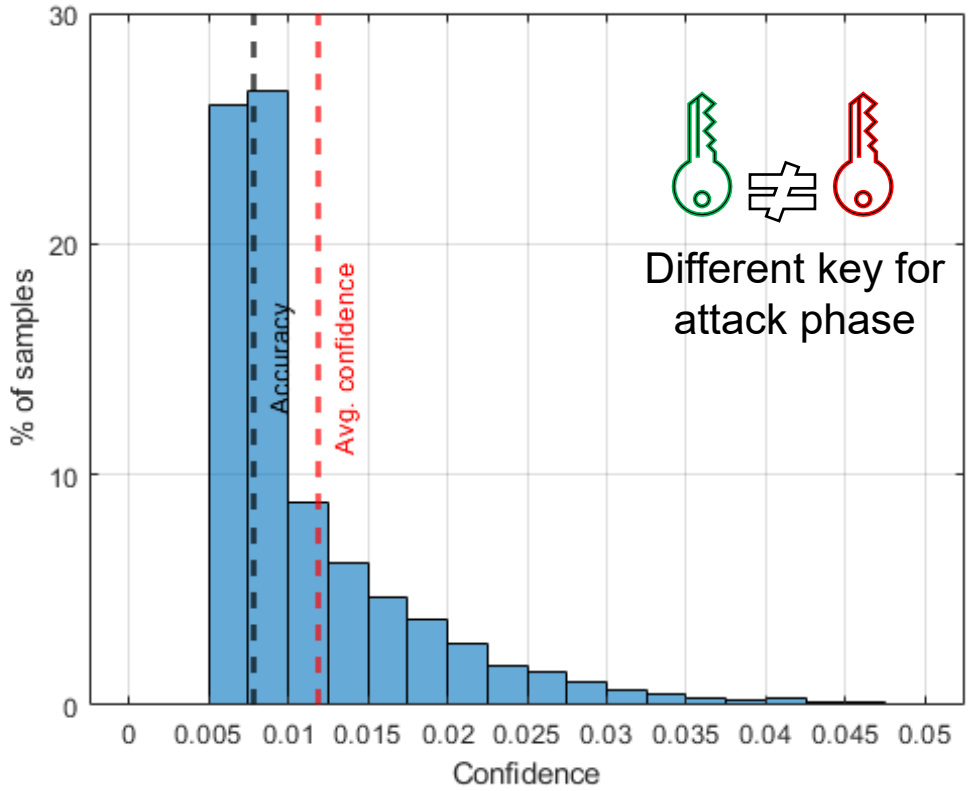


8-bit AVR

AES-128

Masked



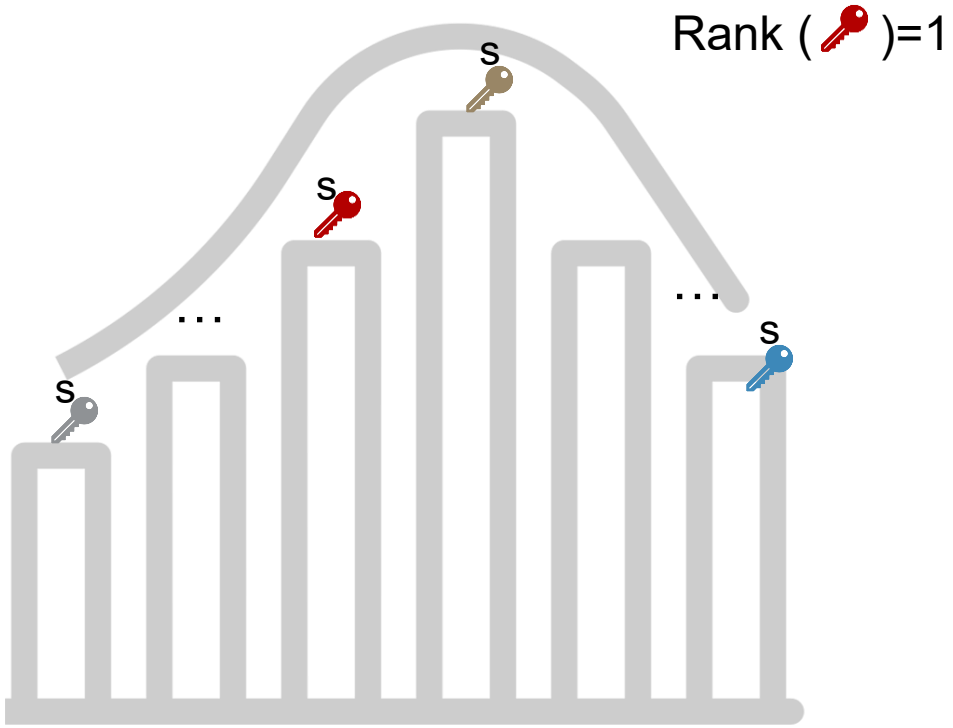






[1] Benadjila, Journal of Cryptographic Engineering, 2020. [2] Wouters, TCHES, 2020. [3] Wu, TETC, 2020.

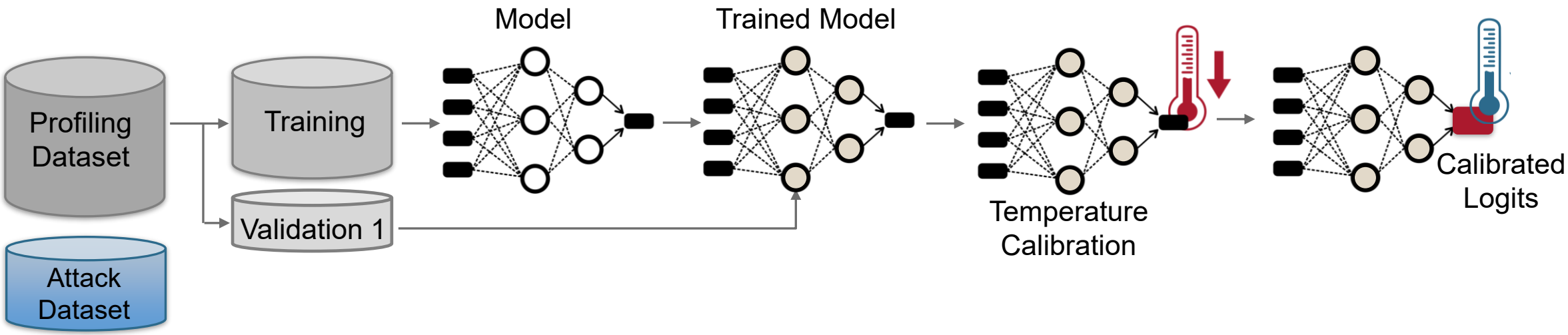
WPI

# Temperature scaling




- Temperature calibration outputs a calibrated probability
  - Take the same raw NN scores, so-called logits
  - Divide them by a temperature T
  - Converting calibrated logits to probabilities
- This makes the output either sharper or softer.
  - T> 1: output becomes softer / less overconfident
  - T < 1: output becomes sharper / more confident.
- How to compute T?!

# Finding T




Training a logistic regression model on the *validation set* to learn a scalar $T > 0$

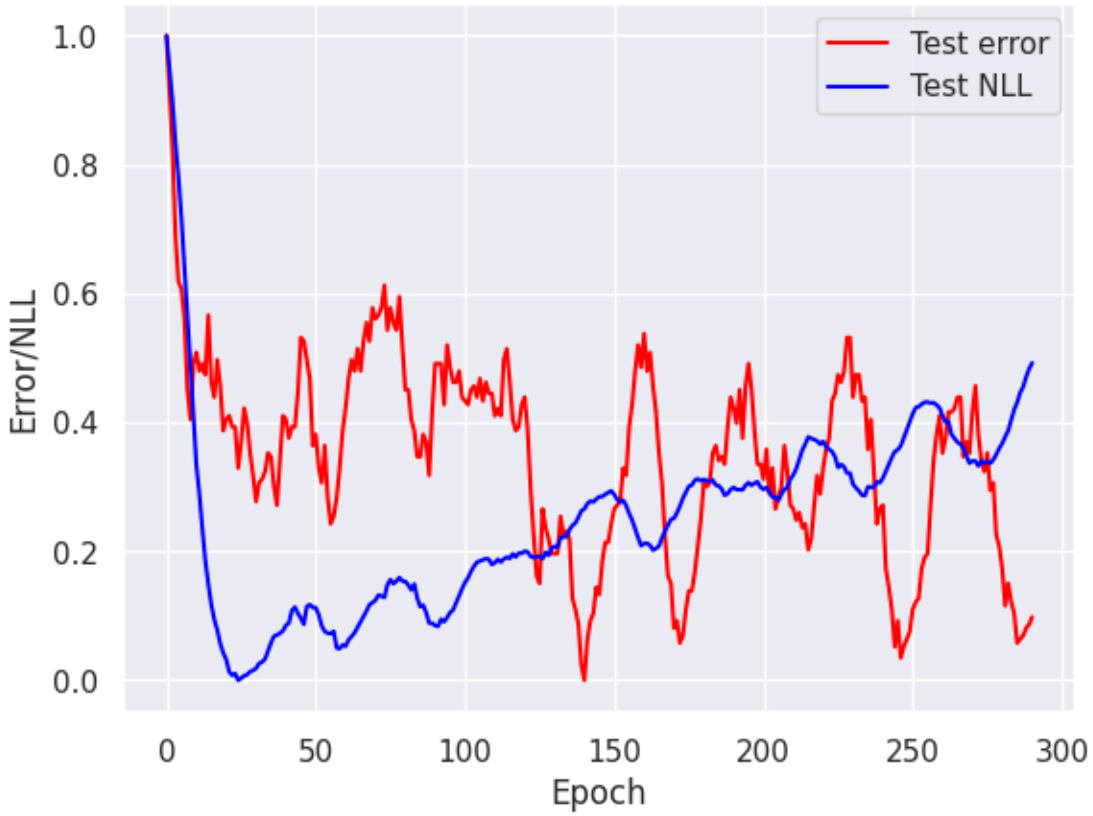
# Some results…

| Dataset | Model | Leakage | T (uncal.) | $TGE_o$ (uncal.) | T (cal.) | $TGE_o$ (cal.) | Std(T) |
|---|---|---|---|---|---|---|---|
| ASCADf | MLP | HW [1] | 6.470 | 3368 | 0.950 | 2736 | 0.09 |
| | CNN | ID [2] | 3.677 | 508 | 0.910 | 442 | 0.24 |
| ASCADr | MLP | HW [1] | 7.976 | 3443 | 0.977 | 2686 | 0.73 |
| | CNN | HW [1] | 1.670 | 1851 | 0.983 | 1615 | 0.097 |
| CHES-CTF | MLP | HW [1] | 15.194 | 4403 | 1.028 | 3381 | 1.70 |

[1] Wu, TETC, 2020. [2] Wouters, TCHES, 2020.

# Why SCA should not push for higher accuracy levels

- SCA goal: key recovery, not top-1 classification
  - Accuracy: predicted class only
  - GE: ranking quality across key hypotheses
- Problem: high accuracy ≠ well-calibrated probabilities
- Overfitting to NLL: better classification, worse probability distribution
  - Consequence: degraded GE despite acceptable accuracy
- Takeaway: monitor calibration, not accuracy alone




Test error and NLL of the MLP model trained on ASCAD-f datasets

# Conclusion

- Temperature scaling: simple post hoc fix
- Design guideline: smaller, better-calibrated networks over larger, overconfident ones
  - Large capacity: lower classification error
    - But also: higher overconfidence risk
    - More epochs: possible NLL overfitting
    - Effect: weaker probability quality for SCA
- Link between temperature scaling and effective perceived information (PI)/cross-entropy (CE) [1,2]
  - "Raw" CE or PI can be misleading as attack-performance predictors
  - Temperature: the source of CE/PI ambiguity
  - Effective CE/PI by optimizing CE (or PI) over $\beta = \frac{1}{T}$
  - A practical way to pick an inverse temperature using the validation dataset, rather than sweeping β abstractly as in [1,2]

[1] Ito, TCHES, 2022. [2] Ito, TCHES, 2025.

# Thank you

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